Foundations of Independent component analysis

[Independent component analysis]

1.0 Content Block

where D is a diagonal matrix with positive entries (assuming X ∗ {\displaystyle X^{*}} has maximum rank), and Q is an orthogonal matrix. Writing the SVD of the mixing matrix A = U Σ V T {\displaystyle A=U\Sigma V^{T}} and comparing with A A T = U Σ 2 U T {\displaystyle AA^{T}=U\Sigma ^{2}U^{T}} the mixing A has the form

2.0 Content Block

h (Y) = 1 N ∑ t = 1 N ln ■ p s (y t) + ln ■ | W | {\displaystyle h(\mathbf {Y})={\frac {1}{N}}\sum _{t=1}^{N}\ln p_{\mathbf {s} }(\mathbf {y} ^{t})+\ln |\mathbf {W} |}

3.0 Content Block

since H (x) = – 1 N ∑ t = 1 N ln ■ p x (x t) {\displaystyle H(\mathbf {x})=-{\frac {1}{N}}\sum _{t=1}^{N}\ln p_{\mathbf {x} }(\mathbf {x} ^{t})} , and maximizing W {\displaystyle \mathbf {W} } does not affect H x {\displaystyle H_{\mathbf {x} }} , so we can maximize the function

4.0 Content Block

Thus, ICA reduces to finding the orthogonal matrix V {\displaystyle V} . This matrix can be computed using optimization techniques via projection pursuit methods (see Projection Pursuit). Well-known algorithms for ICA include infomax, FastICA, JADE, and kernel-independent component analysis, among others. In general, ICA cannot identify the actual number of source signals, a uniquely correct ordering of the source signals, nor the proper scaling (including sign) of the source signals. ICA is important to blind signal separation and has many practical applications. It is closely related to (or even a special case of) the search for a factorial code of the data, i.e., a new vector-valued representation of each data vector such that it gets uniquely encoded by the resulting code vector (loss-free coding), but the code components are statistically independent.

5.0 Content Block

image steganography optical Imaging of neurons neuronal spike sorting face recognition modelling receptive fields of primary visual neurons predicting stock market prices mobile phone communications colour based detection of the ripeness of tomatoes removing artifacts, such as eye blinks, from EEG data. predicting decision-making using EEG analysis of changes in gene expression over time in single cell RNA-sequencing experiments. studies of the resting state network of the brain. astronomy and cosmology finance

6.0 Content Block

A proof can be found in the original papers of Comon; it has been reproduced in the book Independent Component Analysis by Aapo Hyvärinen, Juha Karhunen, and Erkki Oja This approximation also suffers from the same problem as kurtosis (sensitivity to outliers). Other approaches have been developed.

7.0 Content Block

Linear noisy ICA With the added assumption of zero-mean and uncorrelated Gaussian noise n ∼ N (0 , diag ■ (Σ)) {\displaystyle n\sim N(0,\operatorname {diag} (\Sigma))} , the ICA model takes the form x = A s + n {\displaystyle {\boldsymbol {x}}={\boldsymbol {A}}{\boldsymbol {s}}+n} .

8.0 Content Block

Projection pursuit Signal mixtures tend to have Gaussian probability density functions, and source signals tend to have non-Gaussian probability density functions. Each source signal can be extracted from a set of signal mixtures by taking the inner product of a weight vector and those signal mixtures where this inner product provides an orthogonal projection of the signal mixtures. The remaining challenge is finding such a weight vector. One type of method for doing so is projection pursuit. Projection pursuit seeks one projection at a time such that the extracted signal is as non-Gaussian as possible. This contrasts with ICA, which typically extracts M signals simultaneously from M signal mixtures, which requires estimating a M × M unmixing matrix. One practical advantage of projection pursuit over ICA is that fewer than M signals can be extracted if required, where each source signal is extracted from M signal mixtures using an M-element weight vector. We can use kurtosis to recover the multiple source signal by finding the correct weight vectors with the use of projection pursuit. The kurtosis of the probability density function of a signal, for a finite sample, is computed as

9.0 Content Block

Based on infomax Infomax ICA is essentially a multivariate, parallel version of projection pursuit. Whereas projection pursuit extracts a series of signals one at a time from a set of M signal mixtures, ICA extracts M signals in parallel. This tends to make ICA more robust than projection pursuit. The projection pursuit method uses Gram-Schmidt orthogonalization to ensure the independence of the extracted signal, while ICA use infomax and maximum likelihood estimate to ensure the independence of the extracted signal. The Non-Normality of the extracted signal is achieved by assigning an appropriate model, or prior, for the signal. The process of ICA based on infomax in short is: given a set of signal mixtures x {\displaystyle \mathbf {x} } and a set of identical independent model cumulative distribution functions(cdfs) g {\displaystyle g} , we seek the unmixing matrix W {\displaystyle \mathbf {W} } which maximizes the joint entropy of the signals Y = g (y) {\displaystyle \mathbf {Y} =g(\mathbf {y})} , where y = W x {\displaystyle \mathbf {y} =\mathbf {Wx} } are the signals extracted by W {\displaystyle \mathbf {W} } . Given the optimal W {\displaystyle \mathbf {W} } , the signals Y {\displaystyle \mathbf {Y} } have maximum entropy and are therefore independent, which ensures that the extracted signals y = g – 1 (Y) {\displaystyle \mathbf {y} =g^{-1}(\mathbf {Y})} are also independent. g {\displaystyle g} is an invertible function, and is the signal model. Note that if the source signal model probability density function p s {\displaystyle p_{s}} matches the probability density function of the extracted signal p y {\displaystyle p_{\mathbf {y} }} , then maximizing the joint entropy of Y {\displaystyle Y} also maximizes the amount of mutual information between x {\displaystyle \mathbf {x} } and Y {\displaystyle \mathbf {Y} } . For this reason, using entropy to extract independent signals is known as infomax. Consider the entropy of the vector variable Y = g (y) {\displaystyle \mathbf {Y} =g(\mathbf {y})} , where y = W x {\displaystyle \mathbf {y} =\mathbf {Wx} } is the set of signals extracted by the unmixing matrix W {\displaystyle \mathbf {W} } . For a finite set of values sampled from a distribution with pdf p y {\displaystyle p_{\mathbf {y} }} , the entropy of Y {\displaystyle \mathbf {Y} } can be estimated as: